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CS 621 DATA ANALYTICS

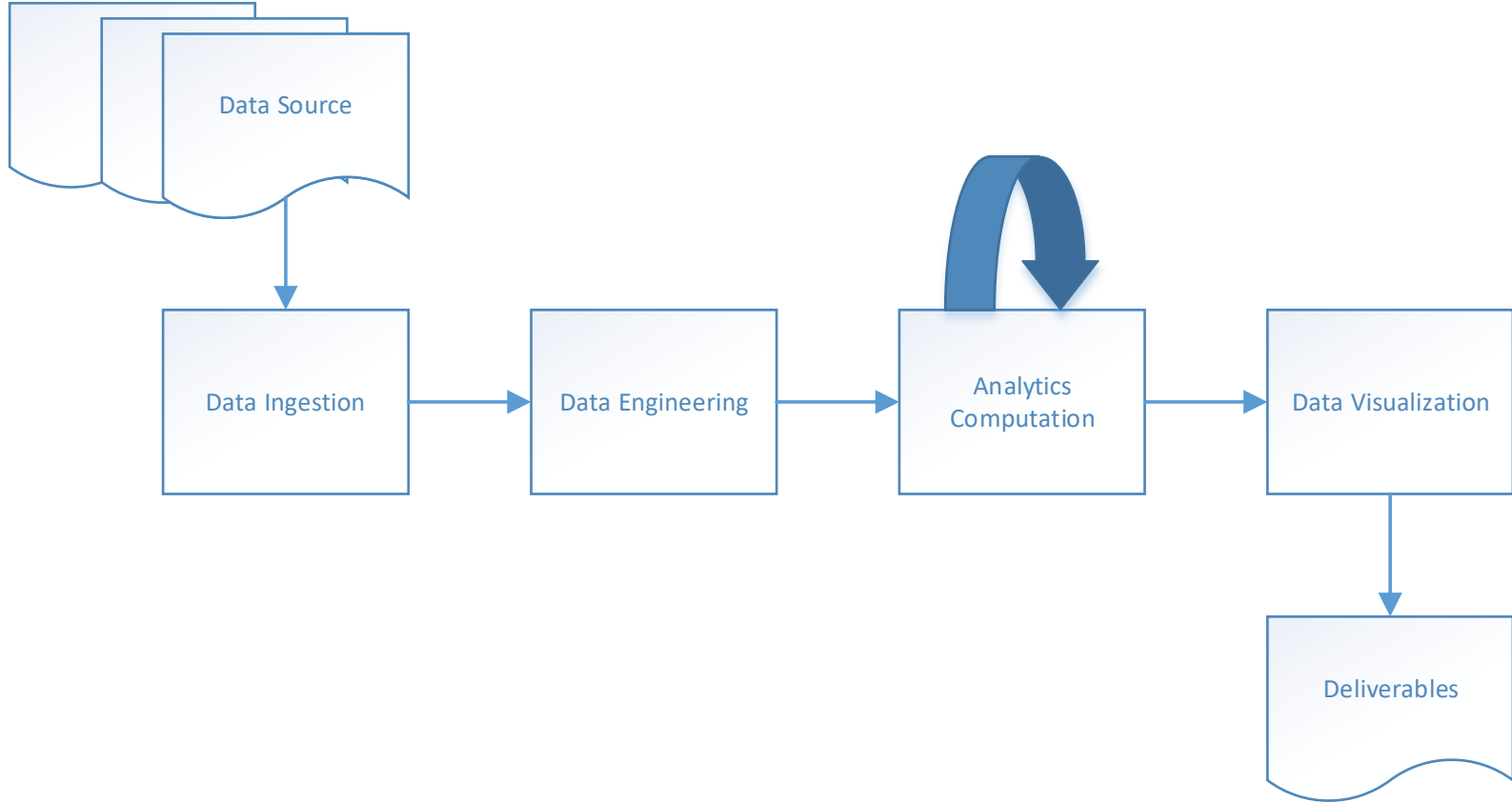
Data Science/Analytics Pipeline

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A TYPICAL DATA SCIENCE PIPELINE



DATA SCIENCE PIPELINE

- A data science pipeline describes the main stages of data analytics process including
 - Data ingestion
 - Data engineering
 - Computing for analytics
 - Visualization



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DATA INGESTION

- Various data sources
- Various data formats
- What data structures will be used to store the data in the software programs?

DATA ENGINEERING

- **Data validation:**
 - Correct data type
 - Missing value vs null value
- **Error checking:**
 - Range of a column, e.g., age of a human object
- **Clean data:**
 - Remove duplicates, remove outliers, etc.

ANALYTICS COMPUTING

- **Basics statistics**
 - Ex: compute average, variance, max, min, median, etc.
- **Compute aggregates**
 - Ex: identify groups
 - Compute desired values for each group
- **Construct models**
 - For example: construct a predictive model to predict your salary in 5 yrs.
- **Machine learning, Data Mining**



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DATA VISUALIZATION

- **Data Visualization will can help us to**
 - explore data
 - understand data
 - demonstrate findings
 - show the future, e.g., future growth prediction.
 - more ...

FINAL DELIVERABLES

- The end result of a pipeline could be
 - a report
 - a software product
 - an input to another pipeline.



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